

$$f(x) = \frac{x^2}{x-2} = x + 2 + \frac{4}{x-2}$$

$$D_f = \mathbb{R} \setminus \{2\}$$

$$\text{Zeros: } f(x)=0 \Leftrightarrow x^2=0 \Leftrightarrow x=0$$

$$\lim_{x \rightarrow \pm\infty} f(x) = \lim_{x \rightarrow \pm\infty} \frac{x^2}{x-2} = \frac{\infty}{\infty} = \lim_{x \rightarrow \pm\infty} \frac{x^2}{x(1-\frac{2}{x})} = \lim_{x \rightarrow \pm\infty} \frac{x}{1-\frac{2}{x}} = \frac{\infty}{1} = \infty$$

$$f'(x) = \left(\frac{x^2}{x-2} \right)' = \frac{(x^2)'(x-2) - (x-2)'x^2}{(x-2)^2} = \frac{2x(x-2) - x^2}{(x-2)^2} = \frac{2x^2 - 4x - x^2}{(x-2)^2} = \frac{x^2 - 4x}{(x-2)^2}$$

$$\text{Zeros: } f'(x) = 0 \Leftrightarrow x^2 - 4x = 0 \Leftrightarrow x(x-4) = 0 \Leftrightarrow x=0 \vee x=4$$

$$f''(x) = \left(\frac{x^2 - 4x}{(x-2)^2} \right)' = \frac{(x^2 - 4x)'(x-2)^2 - ((x-2)^2)'(x^2 - 4x)}{(x-2)^4} = \frac{(2x-4)(x-2)^2 - 2(x-2)(x^2 - 4x)}{(x-2)^4} = \frac{(2x-4)(x-2) - 2(x^2 - 4x)}{(x-2)^3} = \frac{2x^2 - 8x + 8 - 2x^2 + 8x}{(x-2)^3} = \frac{8}{(x-2)^3}$$

$$\text{Zeros: } f''(x) = 0 \Leftrightarrow \text{Não tem zeros}$$

x		0		2		4	
f(x)	-	0	-	~	+		+
f'(x)	+ ↗		- ↘	~	- ↘	0	+ ↗
f''(x)	- ∪		- ∪	~	+ ∩		+ ∩
	↗	Z M	↘	~	↘	m	↗

